DTC P0421:00 [SKYACTIV-D 1.5]

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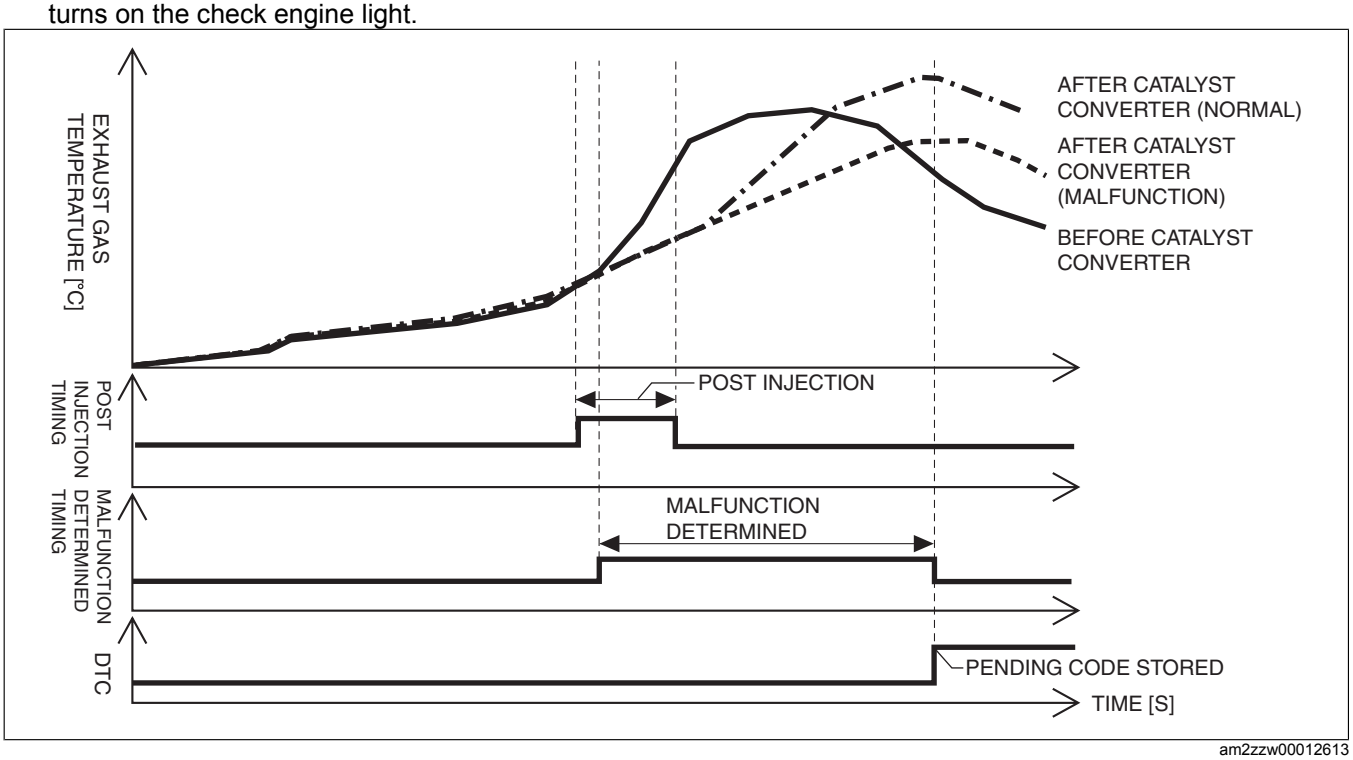
Note
- To determine the malfunctioning part, proceed with the diagnostics from “Function Inspection Using M-MDS”.

Details On DTCs

DESCRIPTION	Catalyst efficiency below threshold	
DETECTION CONDITION	Determination conditions	• The difference between the exhaust gas temperature before and after passing the catalytic converter is specified value or less when the following conditions are met:
	Preconditions	• Battery voltage: **8 V or more** • Engine speed: **1,200—2,000 rpm** • Fuel injection amount: **10—25 mm^3/ stroke** • IAT (IAT sensor No.1): **–10 °C {14 °F} or more** • Barometric pressure: **65 kPa {0.67 kgf/cm^2, 9.5 psi} or more** • ECT: **58 °C {136 °F} or more** • Vehicle speed: **20—140 km/h {13.0—86.9 mph}** • Accumulated PM calculated from fuel: **0.8—4.0 g/l {0.05—0.25 lb/ft^3}** • Accumulated PM calculated from pressure applied to diesel particulate filter: **0.1—2.0 g/l {0.062—0.124 lb/ft^3}** • Traveled distance after diesel particulate filter regeneration: **36—400 km {23—248 mile}** • Exhaust gas temperature before passing catalytic converter: **160—195 °C {320—383 °F}** • Exhaust gas temperature after passing catalytic converter (at engine start): **0—200 °C {32—392 °F}** • Exhaust gas temperature after passing catalytic converter: **135—250 °C {275—482 °F}** • The following DTCs are not detected: — ECT sensor: P0117:00, P0118:00 — IAT sensor No.1: P0112:00, P0113:00 — Fuel pressure sensor: P0192:00, P0193:00 — Exhaust gas temperature sensor No.2: P2032:00, P2033:00 — Exhaust gas temperature sensor No.3: P242C:00, P242D:00 — Glow plug control system: P0670:00, P0671:00, P0672:00, P0673:00, P0674:00 — Fuel injector: P0201:00, P0202:00, P0203:00, P0204:00, P02EE:00, P02EF:00, P02F0:00, P02F1:00, P1378:00, P1379:00, P2146:00, P2147:00, P2148:00, P2149:00, P2150:00, P2151:00
	Drive cycle	• **2**
	Self test type	• CMDTC self test
	Sensor used	• Exhaust gas temperature sensor No.2 • Exhaust gas temperature sensor No.3
FAIL-SAFE FUNCTION	• Not applicable	
VEHICLE STATUS WHEN DTCs ARE OUTPUT	• Check engine light is illuminated	
POSSIBLE CAUSE	• Erratic signal to PCM — Exhaust gas temperature sensor No.2 signal malfunction — Exhaust gas temperature sensor No.3 signal malfunction — Input signal part connector or terminals malfunction — Input signal part related circuit malfunction • Exhaust gas leakage from exhaust system • Exhaust gas temperature sensor No.2 malfunction • Exhaust gas temperature sensor No.3 malfunction • Improper operation of fuel injector • Catalytic converter or diesel particulate filter malfunction (deformation, damage) • PCM malfunction	

System Wiring Diagram
- Not applicable

Function Explanation (DTC Detection Outline)
- The PCM performs diagnosis by monitoring the reaction heat which occurs during the oxidation reaction of the catalytic converter.
- The PCM monitors the exhaust gas temperature before and after the catalytic converter using exhaust gas temperature sensors No.2 and No.3.
- The PCM performs post injection when the preconditions are met during a drive cycle and it effectuates the oxidation reaction of the catalytic converter. At this time, based on the No.2 and No.3 signals, the PCM detects the average and maximum values of each of the exhaust gas temperatures before and after the catalytic converter within the specified time. If the average and maximum exhaust gas temperature values after the catalytic converter are less than the average and maximum exhaust gas temperature values before the catalytic converter, the PCM determines that the performance of catalytic converter has lowered due to deterioration/ damage.
- If the PCM determines a malfunction, it stores a pending code.
- If the PCM determines that the malfunction recurs from the next drive cycle and thereafter, it stores a DTC and turns on the check engine light.



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Repeatability Verification Procedure
1. Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-D 1.5].)
2. Visually inspect the exhaust gas temperature sensor No.2 and No.3 for damage.
3. Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].)
4. Perform compulsory DPF regeneration. (See COMPULSORY DIESEL PARTICULATE FILTER REGENERATION [SKYACTIV-D 1.5].)
5. Access the following PIDs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].)
   — EXHTEMP2
   — EXHTEMP3
6. Drive the vehicle and put PID EXHTEMP2 and EXHTEMP3 each under the following conditions.
   — EXHTEMP2: 300 °C {572 °F} or more
   — EXHTEMP3: 500 °C {932 °F} or more

PID Item/Simulation Item Used In Diagnosis

PID/DATA monitor item table

Item	Definition	Unit	Condition/Specification
EXHTEMP2	Exhaust temperature sensor No.2	°C, °F	• Displays the exhaust gas temperature before passing catalytic converter
EXHTEMP3	Exhaust temperature sensor No.3	°C, °F	• Displays the exhaust gas temperature after passing catalytic converter

Function Inspection Using M-MDS

STEP	INSPECTION		ACTION
1	**PURPOSE: VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
2	**PURPOSE: IDENTIFY TRIGGER DTC FOR FREEZE FRAME DATA** • Is the DTC P0421:00 on FREEZE FRAME DATA?	Yes	Go to the next step.
		No	Go to the troubleshooting procedure for DTC on FREEZE FRAME DATA. (See DTC TABLE [SKYACTIV-D 1.5].)
3	**PURPOSE: RECORD FREEZE FRAME DATA/ SNAPSHOT DATA AND DIAGNOSTIC MONITORING TEST RESULTS TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data and DIAGNOSTIC MONITORING TEST RESULTS (misfire related) on the repair order.	—	Go to the next step.
4	**PURPOSE: VERIFY IF DIAGNOSTIC RESULT IS AFFECTED BY OTHER RELATED DTCs OCCURRING** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the other PENDING CODE/DTC also present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.
5	**PURPOSE: VERIFY IF THERE IS PID ITEM CAUSING DRASTIC CHANGES OF ACCELERATION FLUCTUATION BY INPUT SIGNAL TO PCM AND DSC HU/CM** • Access the following PIDs using the M-MDS: (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) — EXHTEMP2 — EXHTEMP3 • Is there any signal that is far out of specification?	Yes	Inspect the suspected sensor and related wiring harness. • If there is any malfunction: — Repair or replace the malfunctioning part. — Go to the troubleshooting procedure to perform the procedure from Step 5. • If there is no malfunction: — Go to the next step.
		No	Go to the next step.

STEP	INSPECTION		ACTION
6	**PURPOSE: VERIFY CONNECTOR CONNECTIONS** • Access the following PIDs using the M-MDS: (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) **Warning** **• A hot engine and exhaust system can cause severe burns. Turn off the engine and wait until they are cool before removing the exhaust system.** — EXHTEMP2 — EXHTEMP3 • When the following parts are shaken, does the PID value include a PID item which has changed? — Exhaust gas temperature sensor No.2 — Exhaust gas temperature sensor No.3 — PCM	Yes	Repair or replace the applicable connector parts. Go to the troubleshooting procedure to perform the procedure from Step 5.
		No	Go to the troubleshooting procedure to perform the procedure from Step 1.

Troubleshooting Diagnostic Procedure
Intention of troubleshooting procedure

- Step 1
  - — Perform inspection of exhaust gas leakage from exhaust system.
- Step 2—4
  - — Perform inspection of each separate part.
- Step 5—6
  - — Verify that the primary malfunction is resolved and there are no other malfunctions.

Function Inspection Using M-MDS

STEP	INSPECTION		ACTION
1	**PURPOSE: INSPECT EXHAUST SYSTEM FOR LEAKAGE** • Visually inspect for exhaust gas leakage from the exhaust system. • Is there any malfunction?	Yes	Repair or replace the malfunctioning part according to the inspection results, then go to Step 5.
		No	Go to the next step.
2	**PURPOSE: INSPECT EXHAUST GAS TEMPERATURE SENSOR NO.2** • Inspect the exhaust gas temperature sensor No.2. (See EXHAUST GAS TEMPERATURE SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the exhaust gas temperature sensor No.2, then go to Step 5. (See EXHAUST GAS TEMPERATURE SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
3	**PURPOSE: INSPECT EXHAUST GAS TEMPERATURE SENSOR NO.3** • Inspect the exhaust gas temperature sensor No.3. (See EXHAUST GAS TEMPERATURE SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the exhaust gas temperature sensor No.3, then go to the next step. (See EXHAUST GAS TEMPERATURE SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
4	**PURPOSE: INSPECT FOR MALFUNCTION FUEL INJECTOR** • Inspect the fuel injector. (See FUEL INJECTOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the suspect fuel injector, then go to the next step. (See FUEL INJECTOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Catalytic converter or diesel particulate filter can be considered the cause. • Replace the catalytic converter, then go to the next step. (See EXHAUST SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)

STEP	INSPECTION		ACTION
5	**PURPOSE: PERFORM DTC INSPECTION AND VERIFY IF MALFUNCTIONING PART IS PCM** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-D 1.5].) • Implement the repeatability verification procedure. (See Repeatability Verification Procedure.) • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
6	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.